Lab02_Mazo

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Questão 1

Resposta:

Estatísticas suficientes:

n = quantidade de voos válidos; n_atrasados = quantidade de voos com ARRIVAL_DELAY > 10

Percentual de atraso:

Percentual = n_atrasados / n

Questão 2

```r
library(readr)
library(dplyr)
```

```
Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

    filter, lag

Os seguintes objetos são mascarados por 'package:base':

    intersect, setdiff, setequal, union
```

```r
getStats <- function(input, pos) {

  input |>
    filter(AIRLINE %in% c("AA", "DL", "UA", "US")) |>
    filter(
      !is.na(ARRIVAL_DELAY),
      !is.na(DAY),
      !is.na(MONTH),
      !is.na(AIRLINE)
    ) |>
    group_by(DAY, MONTH, AIRLINE) |>
    summarise(
      n = n(),
      n_atrasados = sum(ARRIVAL_DELAY > 10),
      .groups = "drop"
    )
}
```

Questão 3

```r
banco <- read_csv_chunked(
  "flights.csv.zip",
  callback = DataFrameCallback$new(getStats),
  chunk_size = 100000,
  col_types = cols_only(
    AIRLINE = col_character(),
    DAY = col_integer(),
    MONTH = col_integer(),
    ARRIVAL_DELAY = col_double()
  )
)
```

Questão 4:

```r
computeStats <- function(stats_parciais) {

  stats_parciais |>
    group_by(DAY, MONTH, AIRLINE) |>
```

```r
    summarise(
      n = sum(n),
      n_atrasados = sum(n_atrasados),
      .groups = "drop"
    ) |>
    mutate(
      Data = as.Date(
        paste(2015, MONTH, DAY, sep = "-")
      ),
      Perc = n_atrasados / n,
      Cia = AIRLINE
    ) |>
    select(Cia, Data, Perc)
}

stats <- computeStats(banco)
```

Questão 5:

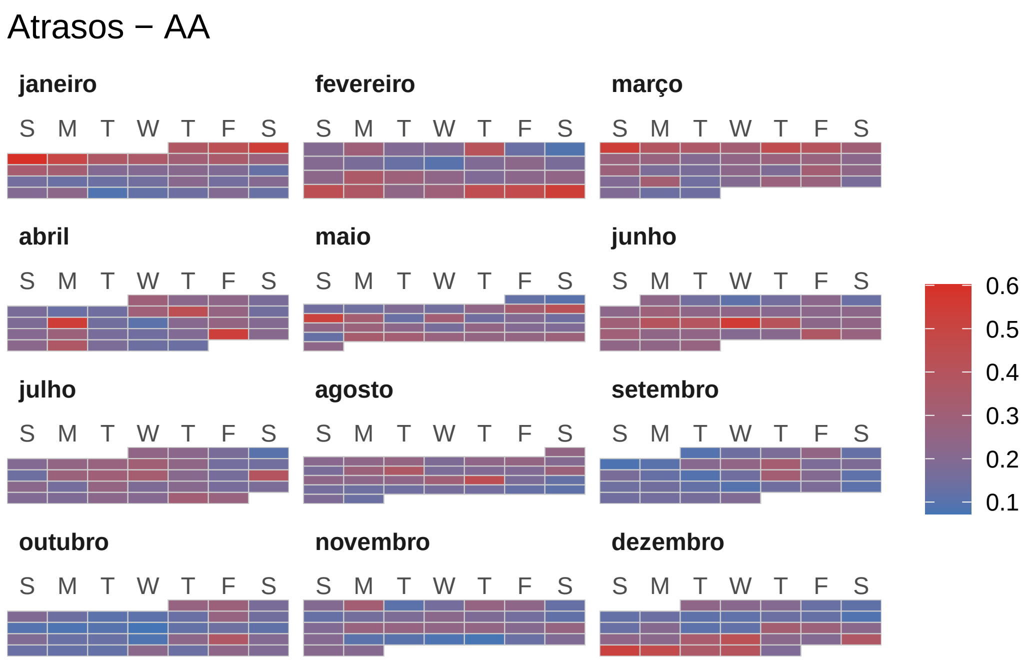
```r
library(devtools)
```

```
Carregando pacotes exigidos: usethis
```

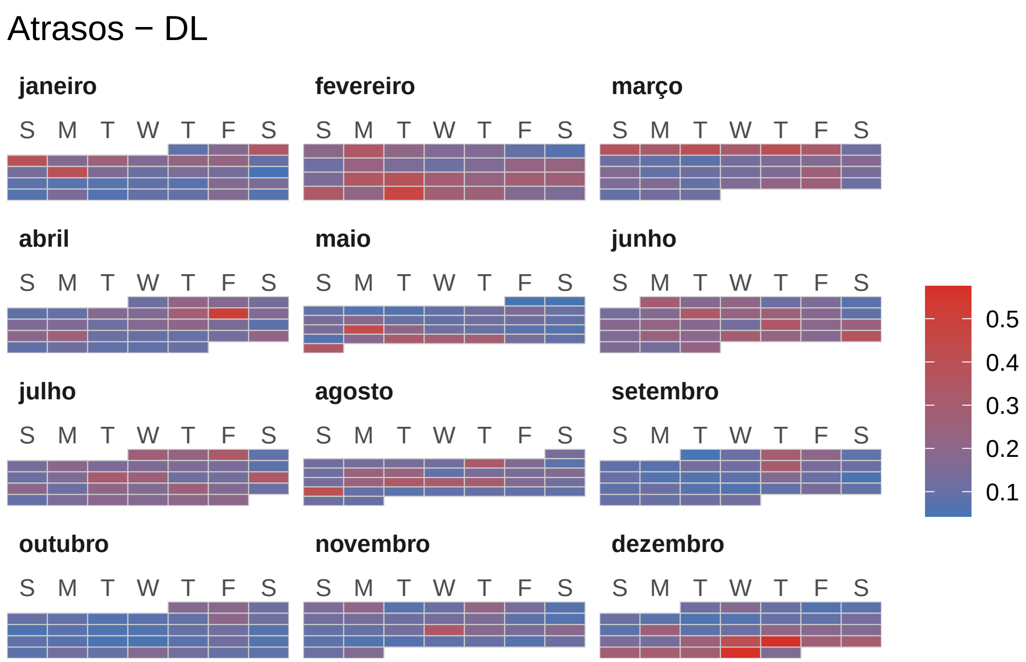
```r
library(ggcal)
library(ggplot2)
```

```r
pal <- scale_fill_gradient(
  low = "#4575b4",
  high = "#d73027"
)

baseCalendario <- function(stats, cia) {

  dados_cia <- stats |>
    filter(Cia == cia)

  ggcal(dados_cia$Data,dados_cia$Perc)
}
```

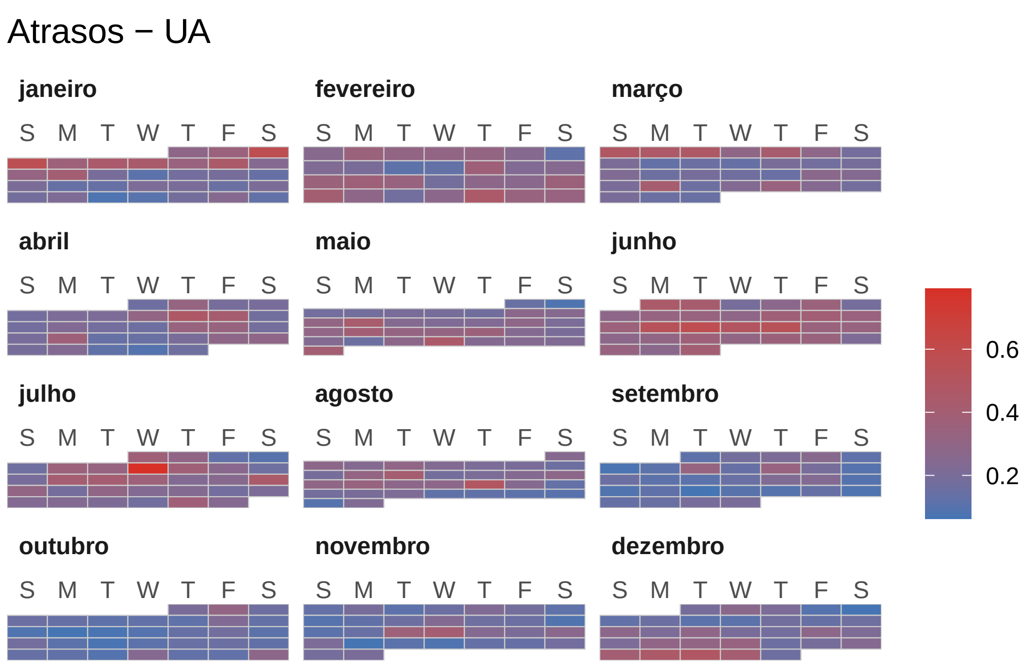
```r
cAA <- baseCalendario(stats, "AA")
cDL <- baseCalendario(stats, "DL")
cUA <- baseCalendario(stats, "UA")
cUS <- baseCalendario(stats, "US")

cAA + pal + ggtitle("Atrasos - AA")
```



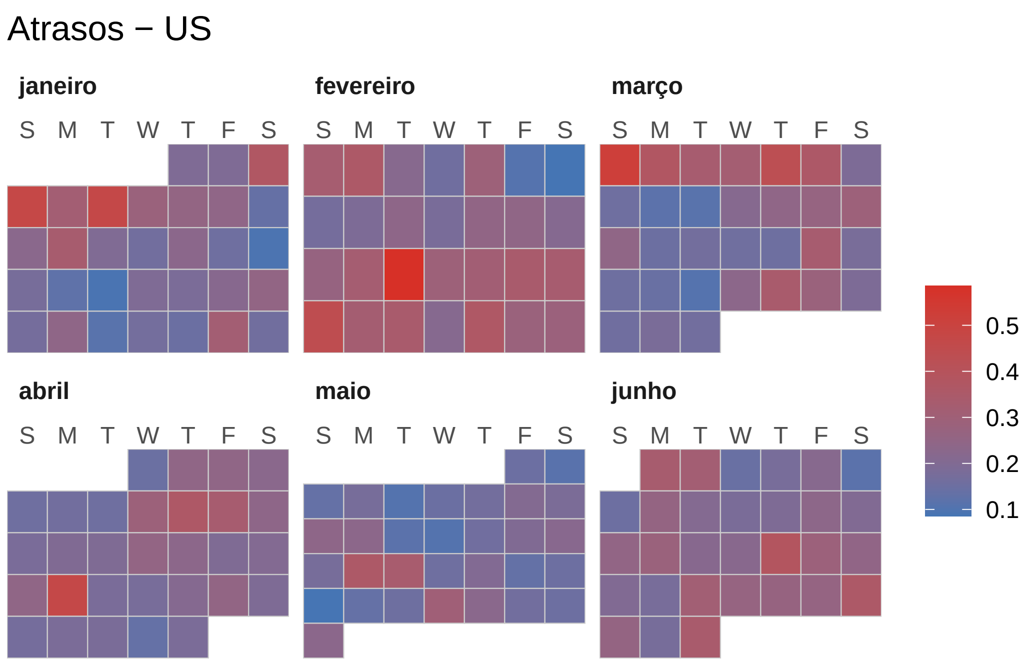

```r
cDL + pal + ggtitle("Atrasos - DL")
```




```r
cUA + pal + ggtitle("Atrasos - UA")
```



```r
cUS + pal + ggtitle("Atrasos - US")
```




```r
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-08-24 19:27:16 -03"
```